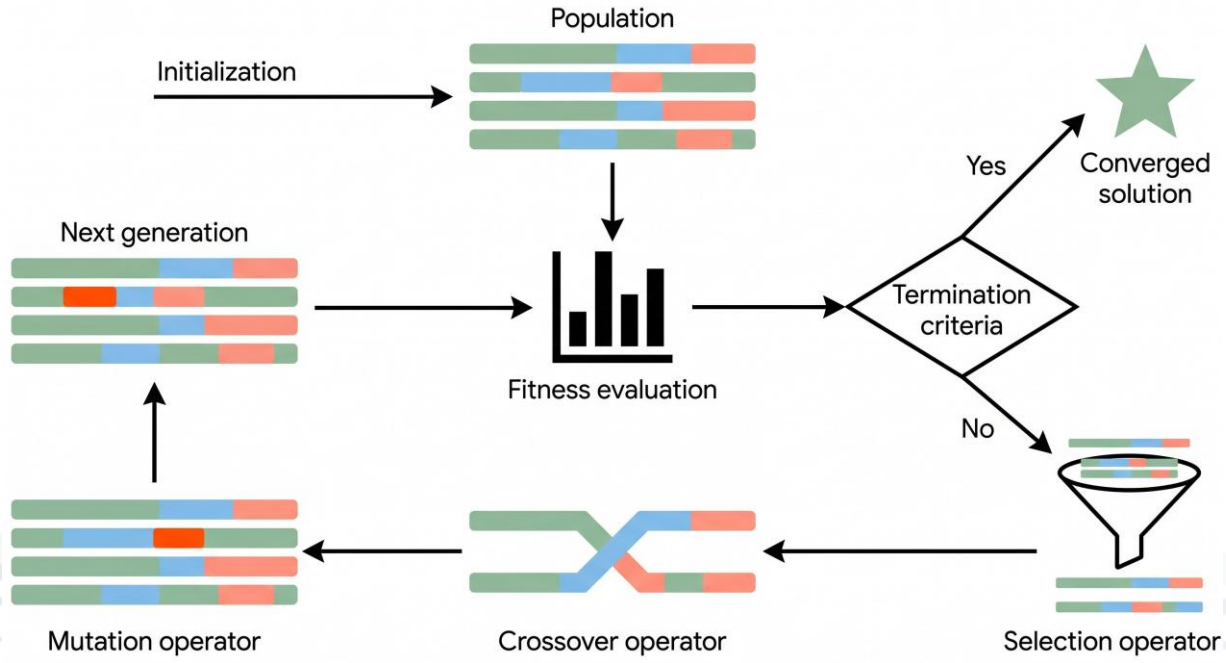
# Discover The Best PN Junction Shape

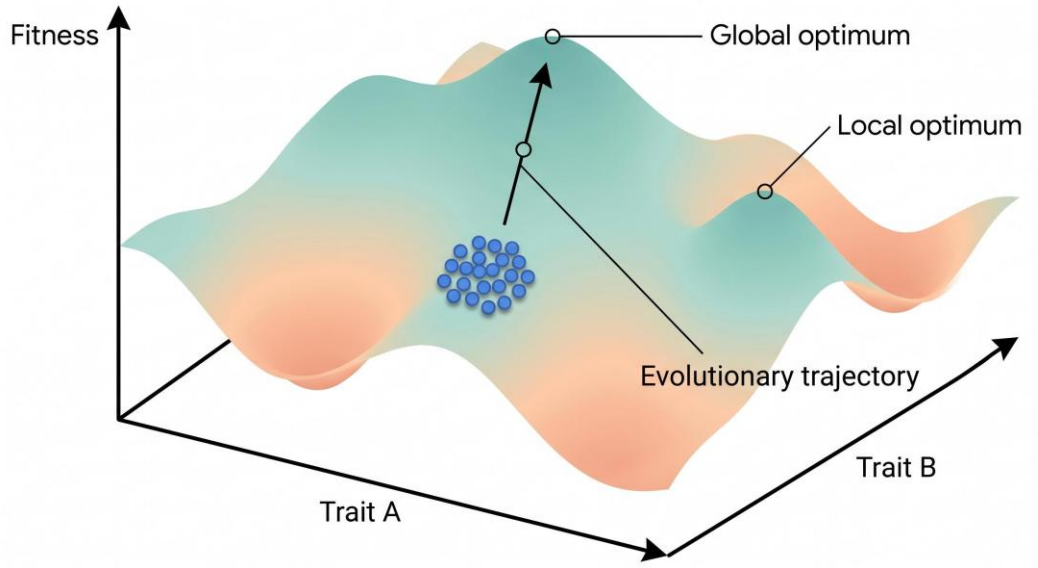
# Genetic Algorithm

- **Initialization:** The algorithm generates an initial "population" of random designs.
- **Fitness Evaluation:** Every design in the population is tested.
- **Selection:** The algorithm selects the best-performing designs from the current generation.
- **Crossover (Recombination):** The selected parents are paired up to "breed." The algorithm takes some design parameters from Parent A and some from Parent B to create an "offspring." The assumption is that combining traits from two good designs might yield an exceptional one
- **Mutation:** To prevent the algorithm from getting stuck in a rut, random tweaks are introduced. A random parameter in an offspring is slightly altered (e.g., increasing a waveguide width by a random few nanometers).

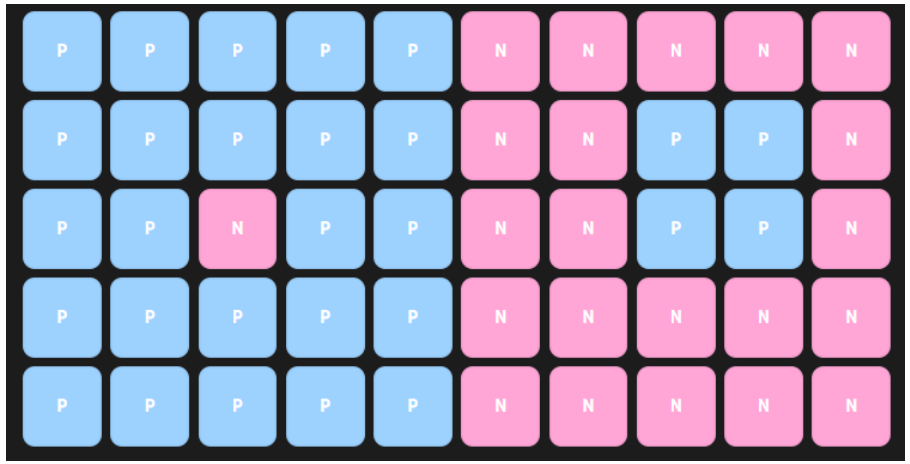


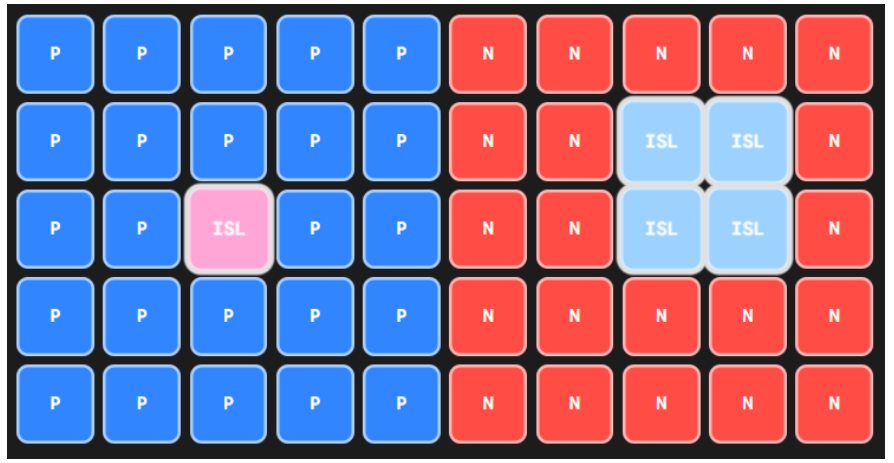
# Escaping Local Minima

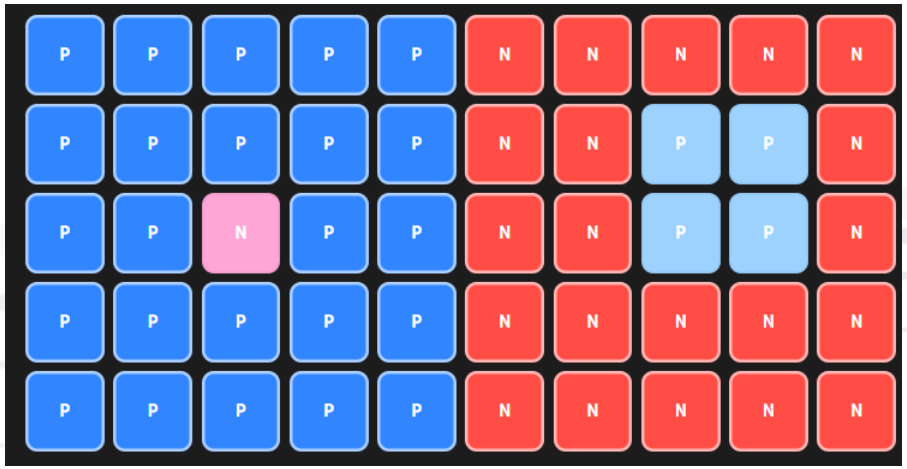
- Standard optimization methods (like gradient descent) are like a blindfolded person walking uphill. They will eventually reach a peak, but if the landscape has many hills, they will stop at the *first* peak they find (a local optimum), never realizing a much higher peak (the global optimum) exists across the valley.
- Because genetic algorithms maintain a diverse population scattered across the entire landscape, and because **mutation** randomly teleports solutions to new areas, they are highly effective at finding the true global optimum

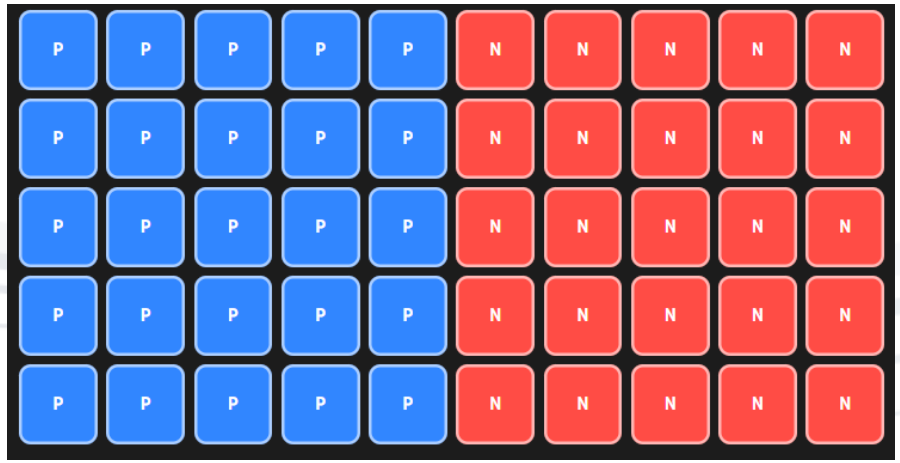


# Repair Function

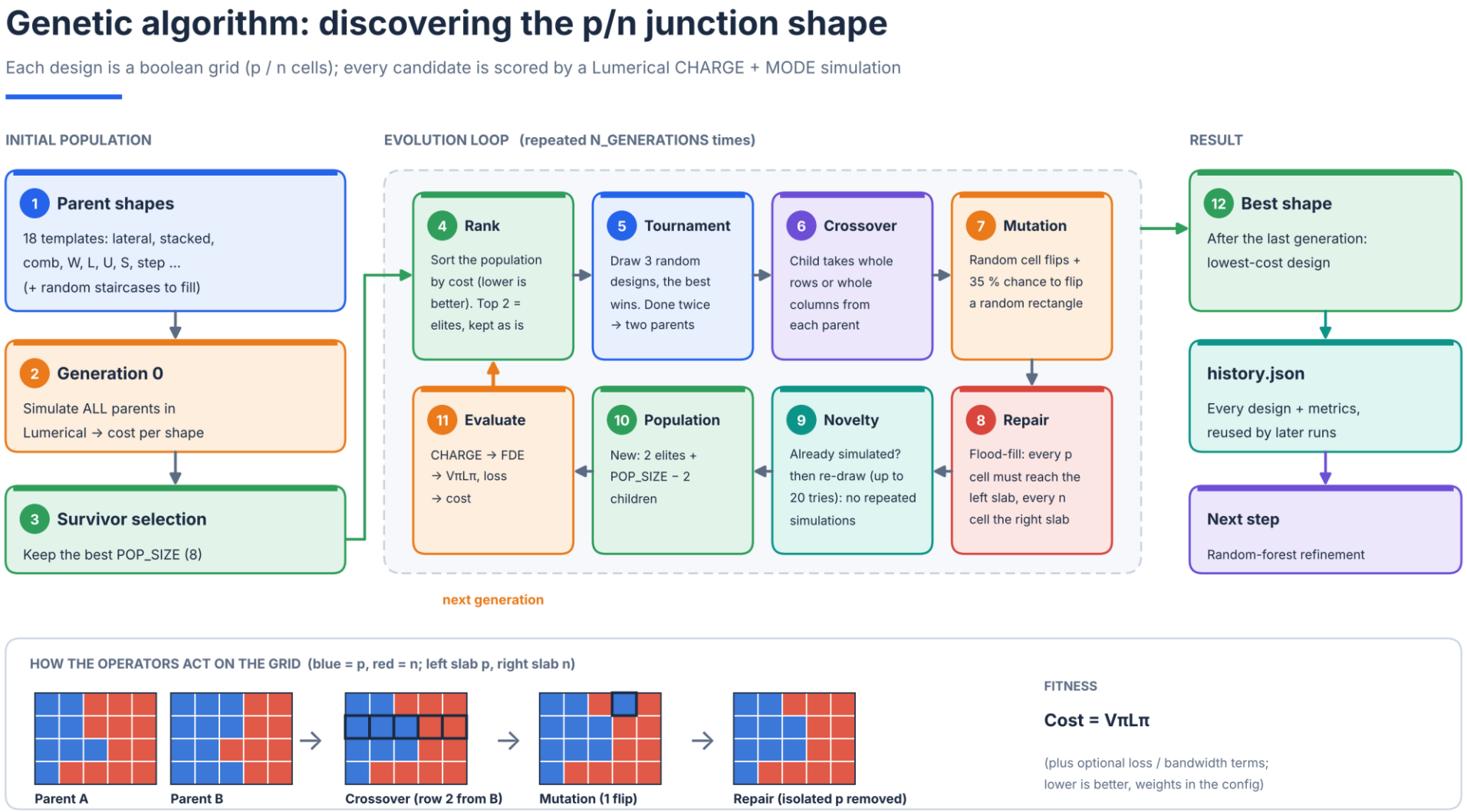


# GA code block diagram

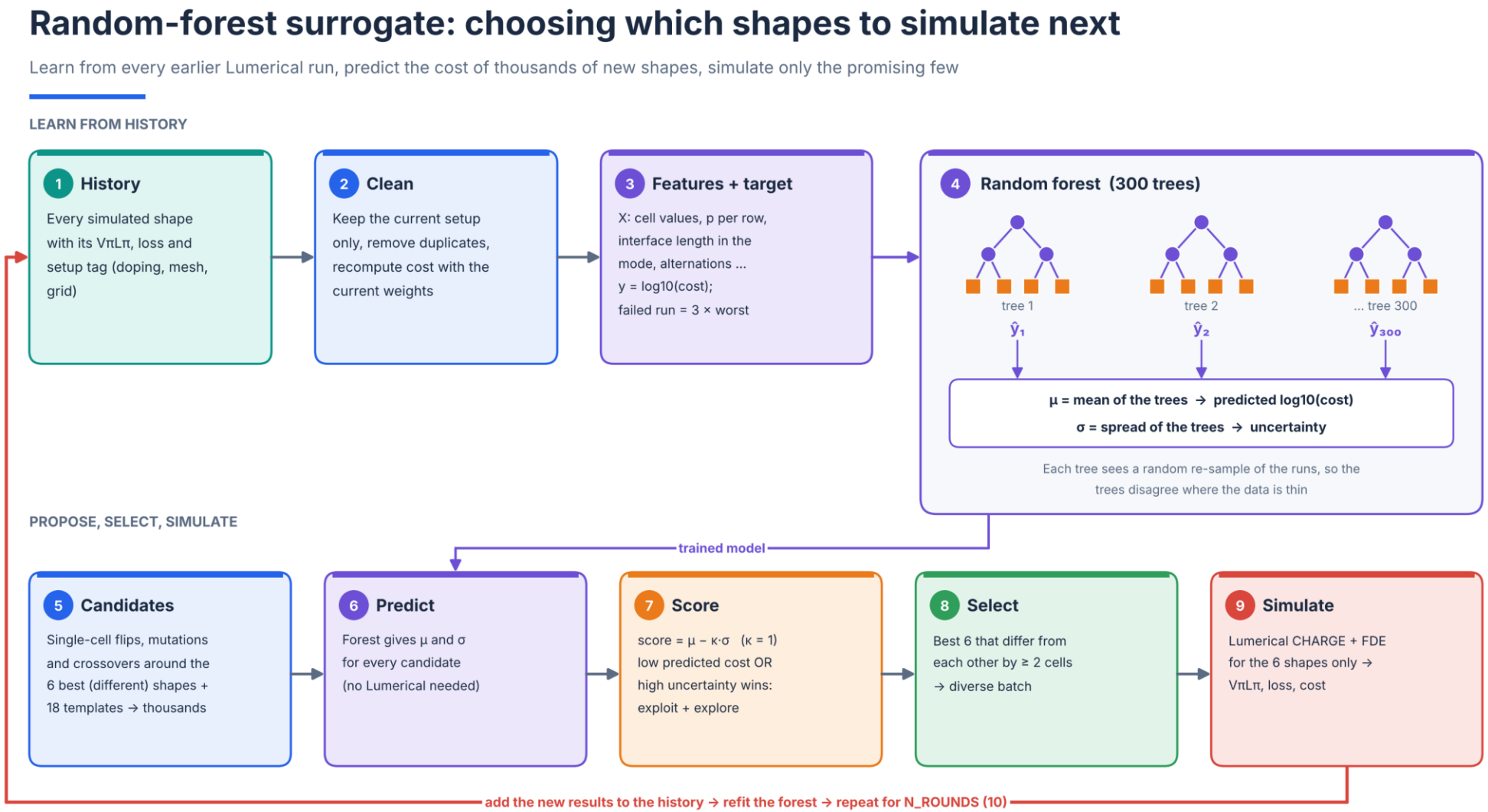


# Random Forest

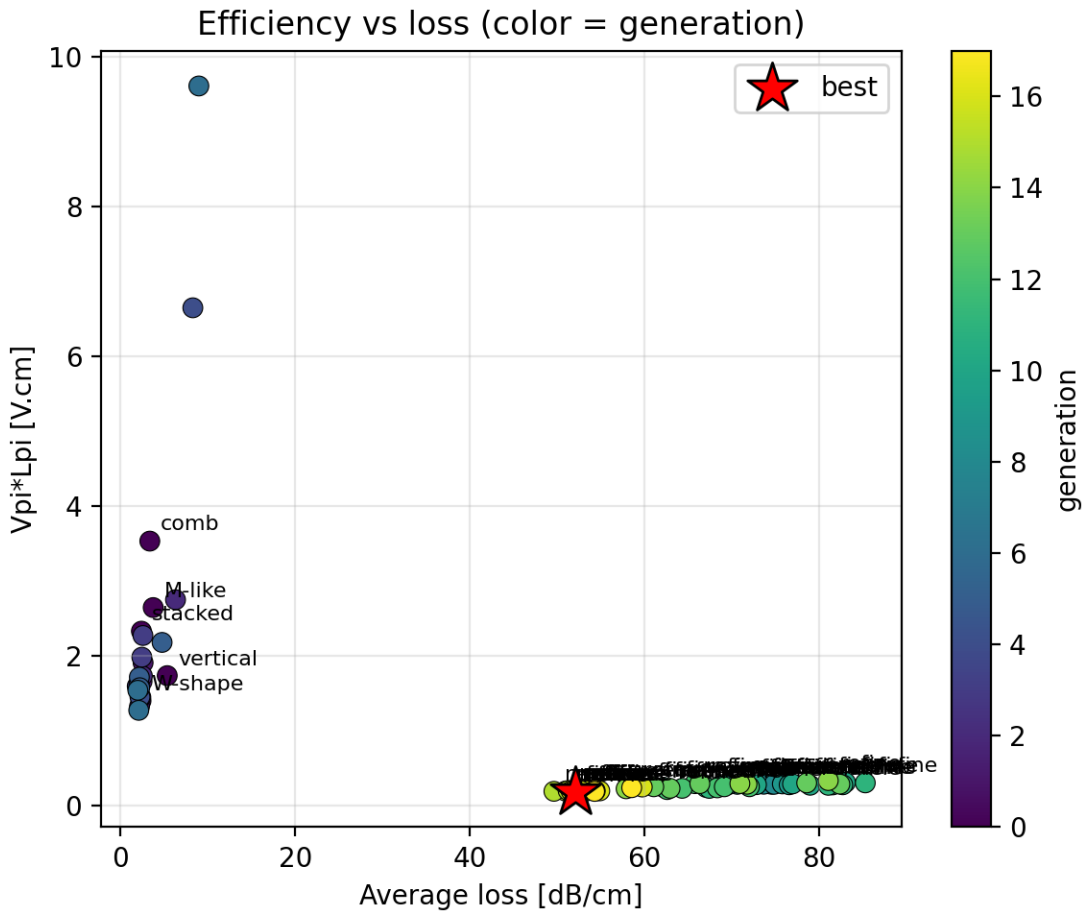
- **Decentralization**
  - Multiple independent decision trees.
- **Randomization**
  - Trained on randomized data subsets.
- **Consensus**
  - Final prediction via majority rule.

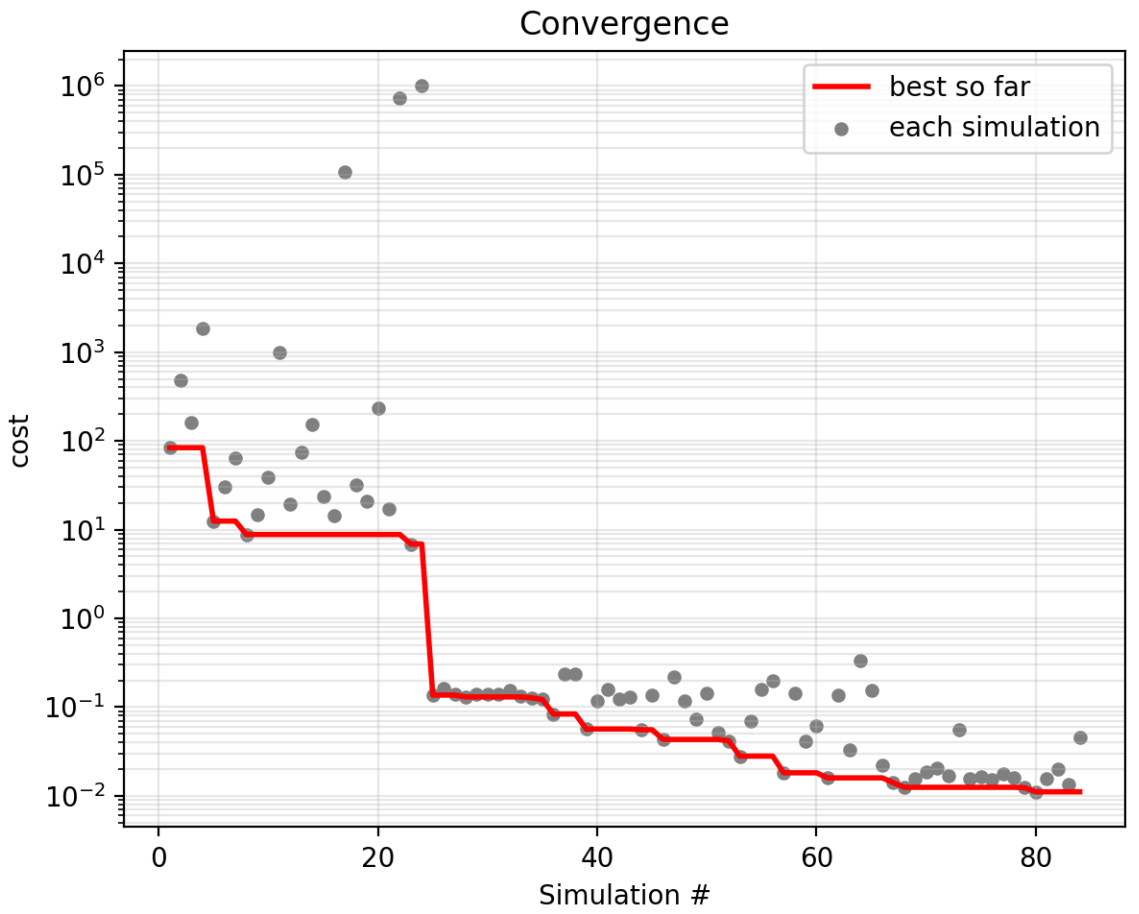


# RF code block diagram

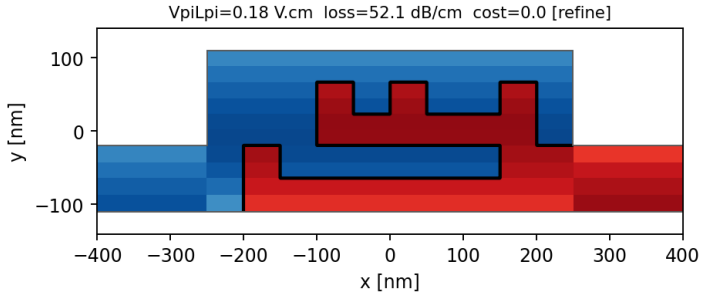


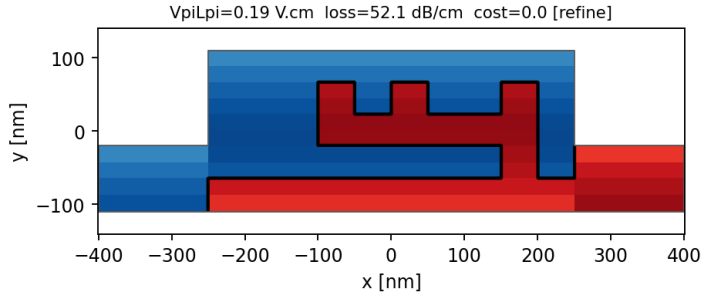
# Results

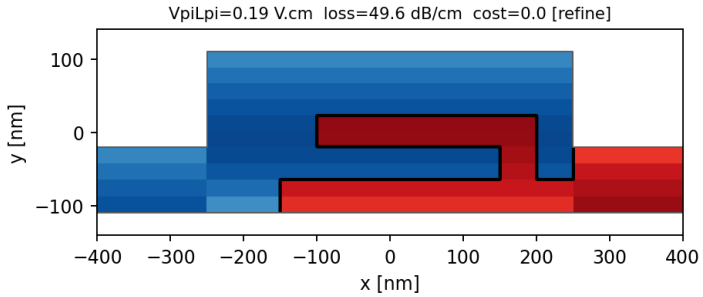

Efficiency vs loss (color = generation)
best
comb
M-like
stacked
vertical
W-shape
Vpi*Lpi [V.cm]
Average loss [dB/cm]
generation
0
2
4
6
8
10
12
14
16
0
20
40
60
80

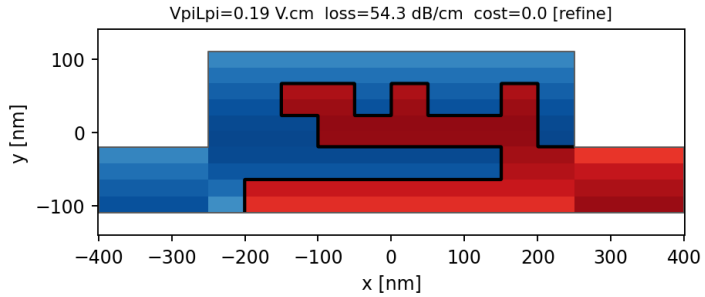

Convergence
best so far
each simulation
cost
Simulation #
$10^{6}$
$10^{5}$
$10^{4}$
$10^{3}$
$10^{2}$
$10^{1}$
$10^{0}$
$10^{-1}$
$10^{-2}$
0
20
40
60
80

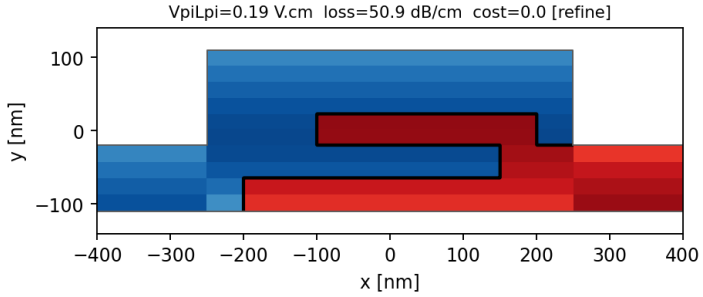
# Results

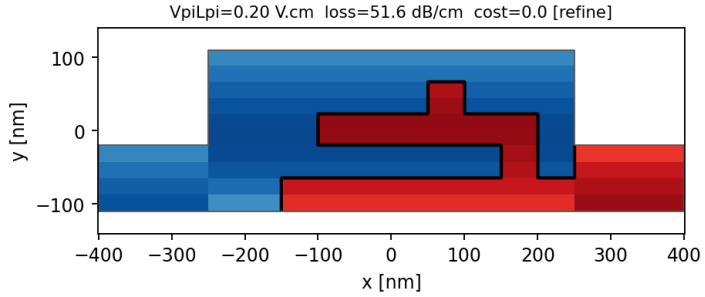

VpiLpi=0.18 V.cm  loss=52.1 dB/cm  cost=0.0 [refine]
y [nm]
x [nm]

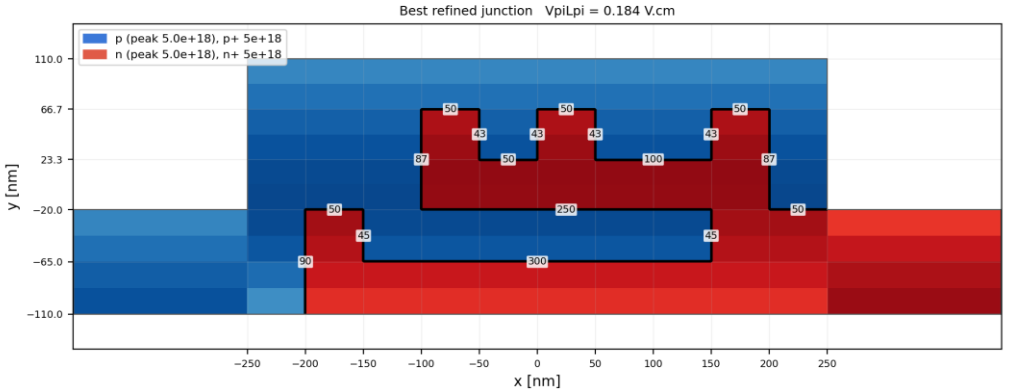

VpiLpi=0.19 V.cm  loss=52.1 dB/cm  cost=0.0 [refine]
y [nm]
x [nm]


VpiLpi=0.19 V.cm  loss=49.6 dB/cm  cost=0.0 [refine]
y [nm]
x [nm]


VpiLpi=0.19 V.cm  loss=54.3 dB/cm  cost=0.0 [refine]
y [nm]
x [nm]


VpiLpi=0.19 V.cm  loss=50.9 dB/cm  cost=0.0 [refine]
y [nm]
x [nm]


VpiLpi=0.20 V.cm  loss=51.6 dB/cm  cost=0.0 [refine]
y [nm]
x [nm]


Best refined junction  VpiLpi = 0.184 V.cm
p (peak 5.0e+18), p+ 5e+18
n (peak 5.0e+18), n+ 5e+18
y [nm]
x [nm]

# Results

- Vpi*Lpi = 0.184 V.cm
- loss(0V) = 65.183, loss(-2V) = 38.965 dB/cm
- cost = 0.011

